

GATE QUESTION Q36

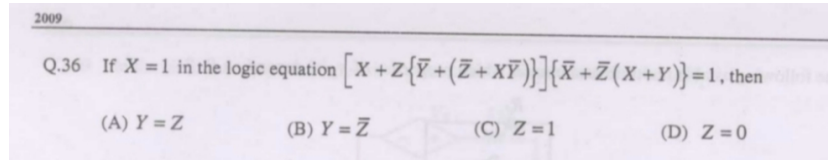


Figure 1: GATE Question Q36

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1 Components

Component	Specification	Quantity
Standard	1 Arduino Uno	1 Breadboard
1 IC 7447	BCD to 7 Segment Decoder	ATmega328P
Common Anode	1 Jumper Wires	1 7 Segment Display
As Required		Male-Male

1.1 Breadboard

The breadboard is used for making temporary circuit connections.

1.2 Arduino Uno

Arduino Uno is used to generate the BCD inputs.

1.3 IC 7447

The 7447 is a BCD to Seven Segment Decoder Driver used with common anode displays.

1.4 Seven Segment Display

The display is used to show the output digit.

2 Boolean Simplification

For $X=1$,

[$F=$]

For output 1,

[$Z=0$]

3 Hardware Connections

3.1 Arduino to 7447

Arduino Pin	7447 Pin
A D3	B D4
C D5	D height

3.2 7447 to Seven Segment Display

7447 Output	Display Segment heighta
a b	b c
c d	d e
e f	f g
g height	

The common anode pin of the display is connected to +5V.

4 Software Implementation

```
#include <Arduino.h>
```

```
void disp_7447(int D,int C,int B,int A)
{
digitalWrite(2,A);
digitalWrite(3,B);
digitalWrite(4,C);
digitalWrite(5,D);
}
```

```
void setup()
{
pinMode(2,OUTPUT);
pinMode(3,OUTPUT);
pinMode(4,OUTPUT);
pinMode(5,OUTPUT);
}
```

```
void loop()
{
int Z = 0;
int F = !Z;

if(F == 1)
disp_7447(0,0,0,1);
else
disp_7447(0,0,0,0);
}
```

5 Output

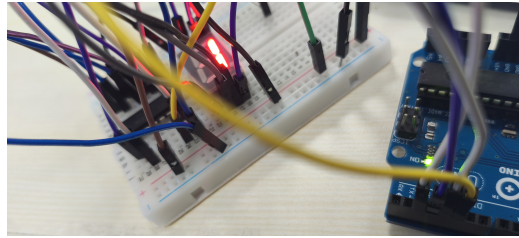


Figure 2: Output Showing 1

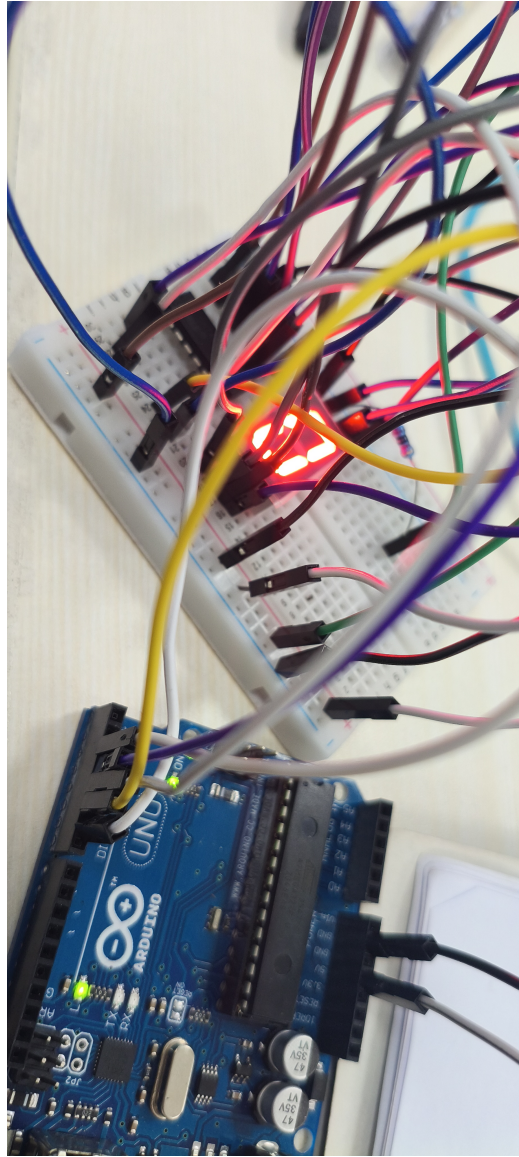


Figure 3: Output Showing 0

6 Result

The given Boolean expression was implemented using Arduino Uno, IC 7447 and a common anode seven segment display.

[F=]

Therefore,

[Z=0]

produces output 1 on the display.